

# CO2mply | Compliance Matrix — Good (87.2%) [development]

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### Item 1

- requirement\_id: R-NK7R-0
- module: project\_data
- title: Protocol eligibility justification
- status: compliant
- risk: low
- score: 100.0
- confidence: 0.5
- project\_evidence:
- methodology\_basis: Module: project\_data | Requirement ID: R-NK7R-0
- gap:
- recommendation: Maintain current evidence and proceed to validation readiness.
- notes: []

### Item 2

- requirement\_id: R-M858-0
- module: project\_data
- title: Legal ownership over removal rights
- status: compliant
- risk: low
- score: 100.0
- confidence: 0.5
- project\_evidence:
- methodology\_basis: Module: project\_data | Requirement ID: R-M858-0
- gap:
- recommendation: Maintain current evidence and proceed to validation readiness.
- notes: []

### Item 3

- requirement\_id: R-7X0X-0
- module: project\_data
- title: Technical description of carbon removal activity
- status: compliant
- risk: low
- score: 100.0
- confidence: 0.5
- project\_evidence:
- methodology\_basis: Module: project\_data | Requirement ID: R-7X0X-0
- gap:
- recommendation: Maintain current evidence and proceed to validation readiness.
- notes: []

### Item 4

- requirement\_id: R-F6R7-0
- module: project\_data
- title: Complete list of project participants
- status: compliant

- risk: low
- score: 100.0
- confidence: 0.5
- project\_evidence:
- methodology\_basis: Module: project\_data | Requirement ID: R-F6R7-0
- gap:
- recommendation: Maintain current evidence and proceed to validation readiness.
- notes: ['[Protocolo] Isometric Standard v1.7, R-F6R7-0 — todos os 7 campos obrigatórios por organização.']

## Item 5

- requirement\_id: R-A5B6-0
- module: project\_data
- title: Project address and/or geo-coordinates
- status: compliant
- risk: low
- score: 100.0
- confidence: 0.5
- project\_evidence:
- methodology\_basis: Module: project\_data | Requirement ID: R-A5B6-0
- gap:
- recommendation: Maintain current evidence and proceed to validation readiness.
- notes: []

## Item 6

- requirement\_id: R-XT6V-0
- module: project\_data
- title: Net carbon removal capacity estimate
- status: partial
- risk: medium
- score: 45.0
- confidence: 0.2
- project\_evidence: Missing fields:
- ghg\_accounting.net\_cdr
- methodology\_basis: Module: project\_data | Requirement ID: R-XT6V-0
- gap: Partial evidence available; some required elements are incomplete.
- recommendation: Provide missing documentation and strengthen evidence for identified gaps.
- notes: ['Estimativa numérica de remoção líquida (tCO2e) não fornecida.', '[Protocolo] Isometric Standard v1.7, R-XT6V-0 — estimativa em tCO2e para o período de crédito completo.']

## Item 7

- requirement\_id: R-VHWJ-0
- module: protocol\_and\_monitoring\_data
- title: Temporal and geographic project boundary defined
- status: compliant
- risk: low
- score: 100.0
- confidence: 0.3333
- project\_evidence:
- methodology\_basis: Module: protocol\_and\_monitoring\_data | Requirement ID: R-VHWJ-0
- gap:
- recommendation: Maintain current evidence and proceed to validation readiness.

- notes: []

## Item 8

- requirement\_id: R-2VKW-0
- module: protocol\_and\_monitoring\_data
- title: System boundary with GHG sources defined
- status: compliant
- risk: low
- score: 100.0
- confidence: 0.3333
- project\_evidence:
- methodology\_basis: Module: protocol\_and\_monitoring\_data | Requirement ID: R-2VKW-0
- gap:
- recommendation: Maintain current evidence and proceed to validation readiness.
- notes: []

## Item 9

- requirement\_id: R-TGBM-0
- module: protocol\_and\_monitoring\_data
- title: GHG statement approach and calculation methodology
- status: compliant
- risk: low
- score: 100.0
- confidence: 0.3333
- project\_evidence:
- methodology\_basis: Module: protocol\_and\_monitoring\_data | Requirement ID: R-TGBM-0
- gap:
- recommendation: Maintain current evidence and proceed to validation readiness.
- notes: []

## Item 10

- requirement\_id: R-PGFH-0
- module: protocol\_and\_monitoring\_data
- title: Baseline scenario reasoned and evidenced
- status: compliant
- risk: low
- score: 100.0
- confidence: 0.5
- project\_evidence:
- methodology\_basis: Module: protocol\_and\_monitoring\_data | Requirement ID: R-PGFH-0
- gap:
- recommendation: Maintain current evidence and proceed to validation readiness.
- notes: []

## Item 11

- requirement\_id: R-HF2G-0
- module: protocol\_and\_monitoring\_data
- title: Leakage assessment provided
- status: partial
- risk: medium
- score: 45.0

- confidence: 0.2
- project\_evidence: Missing fields:
- ghg\_accounting.leakage\_emissions
- methodology\_basis: Module: protocol\_and\_monitoring\_data | Requirement ID: R-HF2G-0
- gap: Partial evidence available; some required elements are incomplete.
- recommendation: Provide missing documentation and strengthen evidence for identified gaps.
- notes: ['Avaliação de leakage não realizada ou não quantificada.', '[Protocolo] Isometric Standard v1.7, R-HF2G-0 — leakage quantificado e deduzido das remoções']

## Item 12

- requirement\_id: R-53Y5-0
- module: protocol\_and\_monitoring\_data
- title: Financial additionality demonstrated
- status: partial
- risk: medium
- score: 60.0
- confidence: 0.3
- project\_evidence: Missing fields:
- eligibility.additionality\_evidence
- methodology\_basis: Module: protocol\_and\_monitoring\_data | Requirement ID: R-53Y5-0
- gap: Partial evidence available; some required elements are incomplete.
- recommendation: Provide missing documentation and strengthen evidence for identified gaps.
- notes: ['Evidência de adicionalidade financeira (revenue principal OU análise IRR) não fornecida.']

## Item 13

- requirement\_id: R-RRST-0
- module: protocol\_and\_monitoring\_data
- title: Common practice additionality demonstrated
- status: compliant
- risk: low
- score: 100.0
- confidence: 1.0
- project\_evidence:
- methodology\_basis: Module: protocol\_and\_monitoring\_data | Requirement ID: R-RRST-0
- gap:
- recommendation: Maintain current evidence and proceed to validation readiness.
- notes: ['[Protocolo] Isometric Standard v1.7, R-RRST-0 — common practice additionality', 'Biochar geralmente abaixo de TRL 8/9 — adicionalidade de prática comum tipicamente automática.']

## Item 14

- requirement\_id: R-CDNF-0
- module: protocol\_and\_monitoring\_data
- title: Environmental additionality (net negative impact) demonstrated
- status: compliant
- risk: low
- score: 100.0
- confidence: 1.0
- project\_evidence:
- methodology\_basis: Module: protocol\_and\_monitoring\_data | Requirement ID: R-CDNF-0
- gap:
- recommendation: Maintain current evidence and proceed to validation readiness.

- notes: ['Impacto climático líquido negativo declarado.']

## Item 15

- requirement\_id: R-983D-0
- module: protocol\_and\_monitoring\_data
- title: Regulatory additionality demonstrated
- status: partial
- risk: medium
- score: 45.0
- confidence: 0.2
- project\_evidence: Missing fields:
- legal.applicable\_environmental\_requirements
- methodology\_basis: Module: protocol\_and\_monitoring\_data | Requirement ID: R-983D-0
- gap: Partial evidence available; some required elements are incomplete.
- recommendation: Provide missing documentation and strengthen evidence for identified gaps.
- notes: ['Requisitos regulatórios aplicáveis não identificados — adicionalidade regulatória não avaliável.', '[Protocolo] Isometric Standard v1.7, R-983D-0 — regulatory additionality']

## Item 16

- requirement\_id: R-KQCS-0
- module: protocol\_and\_monitoring\_data
- title: Regulatory compliance method asserted
- status: partial
- risk: high
- score: 45.0
- confidence: 0.0
- project\_evidence: Missing fields:
- legal.applicable\_environmental\_requirements
- methodology\_basis: Module: protocol\_and\_monitoring\_data | Requirement ID: R-KQCS-0
- gap: Core requirement not met or insufficiently evidenced.
- recommendation: Establish missing core elements required for compliance.
- notes: ['Conformidade com requisitos legais ambientais não evidenciada.', 'Projeto em desenvolvimento: lacuna documental a complementar.']

## Item 17

- requirement\_id: R-7C8E-0
- module: protocol\_and\_monitoring\_data
- title: Durability threshold selected from protocol
- status: compliant
- risk: low
- score: 100.0
- confidence: 0.5
- project\_evidence:
- methodology\_basis: Module: protocol\_and\_monitoring\_data | Requirement ID: R-7C8E-0
- gap:
- recommendation: Maintain current evidence and proceed to validation readiness.
- notes: ['[Protocolo] Isometric Standard v1.7 — R-7C8E-0; Biochar Storage in Soil Environments v1.2 Section 5.1', 'Opção selecionada no PDD: 200 anos.', 'H/Corg e laudos laboratoriais serão avaliados em modo Operacional (R-VGXA-0).']

## Item 18

- requirement\_id: R-1T2Y-0
- module: protocol\_and\_monitoring\_data
- title: Durability in excess of threshold demonstrated
- status: not\_applicable
- risk: none
- score: nan
- confidence: 1.0
- project\_evidence:
- methodology\_basis: Module: protocol\_and\_monitoring\_data | Requirement ID: R-1T2Y-0
- gap:
- recommendation: Providencie esta evidência quando o projeto estiver operacional.
- notes: ['Requisito aplicável apenas em projetos operacionais.', 'Em fase de desenvolvimento/PDD, a evidência operacional ainda não existe — não penaliza o score.']

## Item 19

- requirement\_id: R-F5RZ-0
- module: protocol\_and\_monitoring\_data
- title: Annual average soil temperature method provided (200-year option)
- status: not\_applicable
- risk: none
- score: nan
- confidence: 1.0
- project\_evidence:
- methodology\_basis: Module: protocol\_and\_monitoring\_data | Requirement ID: R-F5RZ-0
- gap:
- recommendation: Providencie esta evidência quando o projeto estiver operacional.
- notes: ['Requisito aplicável apenas em projetos operacionais.', 'Em fase de desenvolvimento/PDD, a evidência operacional ainda não existe — não penaliza o score.']

## Item 20

- requirement\_id: R-V143-0
- module: protocol\_and\_monitoring\_data
- title: Reversal risk assessment completed
- status: compliant
- risk: low
- score: 100.0
- confidence: 0.5
- project\_evidence:
- methodology\_basis: Module: protocol\_and\_monitoring\_data | Requirement ID: R-V143-0
- gap:
- recommendation: Maintain current evidence and proceed to validation readiness.
- notes: ['[Protocolo] Isometric Biochar Storage in Soil Environments v1.2, Section 7.1 — buffer pool 2%', 'Biochar em solo: classificado como Very Low Risk → buffer pool = 2% dos créditos emitidos.']

## Item 21

- requirement\_id: R-2AVD-0
- module: protocol\_and\_monitoring\_data
- title: Sensitivity analysis conducted
- status: partial
- risk: medium
- score: 80.0

- confidence: 0.2667
- project\_evidence: Missing fields:
- quantification.storage\_emissions\_accounted
- methodology\_basis: Module: protocol\_and\_monitoring\_data | Requirement ID: R-2AVD-0
- gap: Partial evidence available; some required elements are incomplete.
- recommendation: Provide missing documentation and strengthen evidence for identified gaps.
- notes: ['Emissões de armazenamento não contabilizadas.']

## Item 22

- requirement\_id: R-K6MA-0
- module: protocol\_and\_monitoring\_data
- title: Uncertainty treatment method specified
- status: partial
- risk: medium
- score: 80.0
- confidence: 0.2667
- project\_evidence: Missing fields:
- quantification.storage\_emissions\_accounted
- methodology\_basis: Module: protocol\_and\_monitoring\_data | Requirement ID: R-K6MA-0
- gap: Partial evidence available; some required elements are incomplete.
- recommendation: Provide missing documentation and strengthen evidence for identified gaps.
- notes: ['Emissões de armazenamento não contabilizadas.']

## Item 23

- requirement\_id: R-Z106-0
- module: protocol\_and\_monitoring\_data
- title: Uncertainty analysis detailed and justified
- status: partial
- risk: medium
- score: 80.0
- confidence: 0.2667
- project\_evidence: Missing fields:
- quantification.storage\_emissions\_accounted
- methodology\_basis: Module: protocol\_and\_monitoring\_data | Requirement ID: R-Z106-0
- gap: Partial evidence available; some required elements are incomplete.
- recommendation: Provide missing documentation and strengthen evidence for identified gaps.
- notes: ['Emissões de armazenamento não contabilizadas.']

## Item 24

- requirement\_id: R-NZQ2-0
- module: protocol\_and\_monitoring\_data
- title: Models described and justified
- status: compliant
- risk: low
- score: 100.0
- confidence: 0.5
- project\_evidence:
- methodology\_basis: Module: protocol\_and\_monitoring\_data | Requirement ID: R-NZQ2-0
- gap:
- recommendation: Maintain current evidence and proceed to validation readiness.
- notes: ['[Protocolo] Isometric Standard v1.7, R-NZQ2-0 — modelos com fonte peer-reviewed, parâmetros e validação', 'Desenvolvimento: verificar se o PDD referencia o modelo Fdurable do Isometric.', 'Equação

$F_{\text{durable},200} = \min(0.95, 1 - [c + (a + b \cdot \ln(T_{\text{soil}})) \cdot H / C_{\text{org}}])$  — parâmetros a/b/c fornecidos pelo Isometric. e temperatura do solo serão avaliados em modo Operacional.']

## Item 25

- requirement\_id: R-GYA1-0
- module: protocol\_and\_monitoring\_data
- title: Data collection and storage approach described
- status: compliant
- risk: low
- score: 100.0
- confidence: 0.5
- project\_evidence:
- methodology\_basis: Module: protocol\_and\_monitoring\_data | Requirement ID: R-GYA1-0
- gap:
- recommendation: Maintain current evidence and proceed to validation readiness.
- notes: ['[Protocolo] Isometric Standard v1.7, R-GYA1-0 — retenção mínima de 5 anos, backup, responsáveis']

## Item 26

- requirement\_id: R-9MJQ-0
- module: environmental\_and\_social\_impact
- title: Environmental regulatory compliance outlined
- status: compliant
- risk: low
- score: 100.0
- confidence: 0.5
- project\_evidence:
- methodology\_basis: Module: environmental\_and\_social\_impact | Requirement ID: R-9MJQ-0
- gap:
- recommendation: Maintain current evidence and proceed to validation readiness.
- notes: []

## Item 27

- requirement\_id: R-X9EC-0
- module: environmental\_and\_social\_impact
- title: Environmental and social impact assessment
- status: compliant
- risk: low
- score: 100.0
- confidence: 0.5
- project\_evidence:
- methodology\_basis: Module: environmental\_and\_social\_impact | Requirement ID: R-X9EC-0
- gap:
- recommendation: Maintain current evidence and proceed to validation readiness.
- notes: []

## Item 28

- requirement\_id: R-4K5P-0
- module: environmental\_and\_social\_impact
- title: No net environmental harm demonstrated
- status: compliant
- risk: low



- score: 100.0
- confidence: 0.5
- project\_evidence:
- methodology\_basis: Module: environmental\_and\_social\_impact | Requirement ID: R-4K5P-0
- gap:
- recommendation: Maintain current evidence and proceed to validation readiness.
- notes: []

## Item 29

- requirement\_id: R-R81B-0
- module: environmental\_and\_social\_impact
- title: No net social harm demonstrated
- status: compliant
- risk: low
- score: 100.0
- confidence: 0.5
- project\_evidence:
- methodology\_basis: Module: environmental\_and\_social\_impact | Requirement ID: R-R81B-0
- gap:
- recommendation: Maintain current evidence and proceed to validation readiness.
- notes: []

## Item 30

- requirement\_id: R-5KQC-0
- module: environmental\_and\_social\_impact
- title: Agricultural productivity and soil quality monitoring documented
- status: not\_applicable
- risk: none
- score: nan
- confidence: 1.0
- project\_evidence:
- methodology\_basis: Module: environmental\_and\_social\_impact | Requirement ID: R-5KQC-0
- gap:
- recommendation: Providencie esta evidência quando o projeto estiver operacional.
- notes: ['Requisito aplicável apenas em projetos operacionais.', 'Em fase de desenvolvimento/PDD, a evidência operacional ainda não existe — não penaliza o score.']

## Item 31

- requirement\_id: R-BWX0-0
- module: environmental\_and\_social\_impact
- title: Alignment with relevant SDGs demonstrated
- status: compliant
- risk: low
- score: 100.0
- confidence: 0.5
- project\_evidence:
- methodology\_basis: Module: environmental\_and\_social\_impact | Requirement ID: R-BWX0-0
- gap:
- recommendation: Maintain current evidence and proceed to validation readiness.
- notes: ['[Protocolo] Isometric Standard v1.7, R-BWX0-0 — ODS relevantes: ODS 13 (Ação Climática), ODS 2 (Fome Zero), ODS 15 (Vida Terrestre).']

## Item 32

- requirement\_id: R-6VFZ-0
- module: environmental\_and\_social\_impact
- title: Project closure conditions and plan described
- status: compliant
- risk: low
- score: 100.0
- confidence: 0.5
- project\_evidence:
- methodology\_basis: Module: environmental\_and\_social\_impact | Requirement ID: R-6VFZ-0
- gap:
- recommendation: Maintain current evidence and proceed to validation readiness.
- notes: '[[Protocolo] Isometric Standard v1.7, R-6VFZ-0 — condições de encerramento e ações pós-cessação.]'

## Item 33

- requirement\_id: R-BC4H-0
- module: environmental\_and\_social\_impact
- title: Adaptive management plan: information sharing, emergency response, pause/stop conditions
- status: compliant
- risk: low
- score: 100.0
- confidence: 0.25
- project\_evidence:
- methodology\_basis: Module: environmental\_and\_social\_impact | Requirement ID: R-BC4H-0
- gap:
- recommendation: Maintain current evidence and proceed to validation readiness.
- notes: '[[Protocolo] Isometric Biochar Production and Storage v1.2, Section 19.2 (R-BC4H-0)', 'Os 4 gatilhos obrigatórios de pausa/stop: (1) falha de instrumentos, (2) poluentes > threshold, (3) não conformidade regulatória, (4) risco à saúde.]'

## Item 34

- requirement\_id: R-MY64-0
- module: environmental\_and\_social\_impact
- title: Pollution prevention: PAHs, heavy metals, PCBs/Dioxins
- status: partial
- risk: medium
- score: 45.0
- confidence: 0.1
- project\_evidence: Missing fields:
- biochar.characterization.pollutants.PCBs
- biochar.characterization.pollutants.dioxins

Failed checks:

- safeguards.mitigation\_plan
- methodology\_basis: Module: environmental\_and\_social\_impact | Requirement ID: R-MY64-0
- gap: Partial evidence available; some required elements are incomplete.
- recommendation: Provide missing documentation and strengthen evidence for identified gaps.
- notes: 'Plano de mitigação de poluentes não documentado.', 'Concentração de PCBs não declarada.', 'Concentração de PCDD/F não declarada.', '[[Protocolo] Isometric Biochar Production and Storage v1.2, Sections 6-7; World Biochar Certificate limits]'

### Item 35

- requirement\_id: R-M760-0
- module: environmental\_and\_social\_impact
- title: Baseline soil samples collected prior to biochar spreading
- status: not\_applicable
- risk: none
- score: nan
- confidence: 1.0
- project\_evidence:
- methodology\_basis: Module: environmental\_and\_social\_impact | Requirement ID: R-M760-0
- gap:
- recommendation: Providencie esta evidência quando o projeto estiver operacional.
- notes: ['Requisito aplicável apenas em projetos operacionais.', 'Em fase de desenvolvimento/PDD, a evidência operacional ainda não existe — não penaliza o score.']

### Item 36

- requirement\_id: R-1YC3-0
- module: environmental\_and\_social\_impact
- title: Co-benefits related to soil health reported (optional)
- status: not\_applicable
- risk: none
- score: nan
- confidence: 1.0
- project\_evidence:
- methodology\_basis: Module: environmental\_and\_social\_impact | Requirement ID: R-1YC3-0
- gap:
- recommendation: Providencie esta evidência quando o projeto estiver operacional.
- notes: ['Requisito aplicável apenas em projetos operacionais.', 'Em fase de desenvolvimento/PDD, a evidência operacional ainda não existe — não penaliza o score.']

### Item 37

- requirement\_id: R-ZHRN-0
- module: stakeholder\_input\_process
- title: Stakeholder consultation documented
- status: partial
- risk: medium
- score: 45.0
- confidence: 0.2
- project\_evidence:
- methodology\_basis: Module: stakeholder\_input\_process | Requirement ID: R-ZHRN-0
- gap: Partial evidence available; some required elements are incomplete.
- recommendation: Strengthen consistency and completeness of existing evidence.
- notes: ['Processo de consulta a stakeholders não documentado (convites, comentários recebidos, ações tomadas).', '[Protocolo] Isometric Standard v1.7, R-ZHRN-0/R-E579-0 — consulta iterativa, acessível, transparente', 'Prazos obrigatórios: reconhecimento de reclamações ≤ 14 dias; resolução ou escalação ≤ 60 dias.']

### Item 38

- requirement\_id: R-E579-0
- module: stakeholder\_input\_process

- title: Grievance mechanism outlined
- status: partial
- risk: medium
- score: 45.0
- confidence: 0.2
- project\_evidence:
- methodology\_basis: Module: stakeholder\_input\_process | Requirement ID: R-E579-0
- gap: Partial evidence available; some required elements are incomplete.
- recommendation: Strengthen consistency and completeness of existing evidence.
- notes: ['Processo de consulta a stakeholders não documentado (convites, comentários recebidos, ações tomadas).', '[Protocolo] Isometric Standard v1.7, R-ZHRN-0/R-E579-0 — consulta iterativa, acessível, transparente', 'Prazos obrigatórios: reconhecimento de reclamações ≤ 14 dias; resolução ou escalção ≤ 60 dias.']

## Item 39

- requirement\_id: R-ENZR-0
- module: appendix
- title: Monitoring parameter table provided
- status: compliant
- risk: low
- score: 100.0
- confidence: 0.5
- project\_evidence:
- methodology\_basis: Module: appendix | Requirement ID: R-ENZR-0
- gap:
- recommendation: Maintain current evidence and proceed to validation readiness.
- notes: ['[Protocolo] Isometric Standard v1.7, R-ENZR-0 — tabela de parâmetros monitorados (fonte, frequência, QA/QC)', 'Tabela deve cobrir: parâmetro, fonte de dados, frequência de medição, procedimentos QA/QC e evidências planejadas.']

## Item 40

- requirement\_id: R-6AQG-0
- module: appendix
- title: Engineering design diagram of pyrolysis reactor
- status: compliant
- risk: low
- score: 100.0
- confidence: 0.25
- project\_evidence:
- methodology\_basis: Module: appendix | Requirement ID: R-6AQG-0
- gap:
- recommendation: Maintain current evidence and proceed to validation readiness.
- notes: ['[Protocolo] Isometric Biochar Production and Storage v1.2, Reactor Design Requirements', 'Desenvolvimento: diagrama e plano de manutenção obrigatórios no PDD.']

## Item 41

- requirement\_id: R-SZK5-0
- module: appendix
- title: Pyrolysis gas leakage sensors described
- status: compliant
- risk: low

- score: 100.0
- confidence: 0.25
- project\_evidence:
- methodology\_basis: Module: appendix | Requirement ID: R-SZK5-0
- gap:
- recommendation: Maintain current evidence and proceed to validation readiness.
- notes: ['[Protocolo] Isometric Biochar Production and Storage v1.2, Reactor Design Requirements', 'Desenvolvimento: diagrama e plano de manutenção obrigatórios no PDD.']

## Item 42

- requirement\_id: R-DMET-0
- module: appendix
- title: Reactor material selection justified
- status: compliant
- risk: low
- score: 100.0
- confidence: 0.25
- project\_evidence:
- methodology\_basis: Module: appendix | Requirement ID: R-DMET-0
- gap:
- recommendation: Maintain current evidence and proceed to validation readiness.
- notes: ['[Protocolo] Isometric Biochar Production and Storage v1.2, Reactor Design Requirements', 'Desenvolvimento: diagrama e plano de manutenção obrigatórios no PDD.']

## Item 43

- requirement\_id: R-19AF-0
- module: appendix
- title: Reactor maintenance plan evidenced
- status: compliant
- risk: low
- score: 100.0
- confidence: 0.25
- project\_evidence:
- methodology\_basis: Module: appendix | Requirement ID: R-19AF-0
- gap:
- recommendation: Maintain current evidence and proceed to validation readiness.
- notes: ['[Protocolo] Isometric Biochar Production and Storage v1.2, Reactor Design Requirements', 'Desenvolvimento: diagrama e plano de manutenção obrigatórios no PDD.']

## Item 44

- requirement\_id: R-S8K1-1
- module: appendix
- title: Sampling procedure described and justified
- status: not\_applicable
- risk: none
- score: nan
- confidence: 1.0
- project\_evidence:
- methodology\_basis: Module: appendix | Requirement ID: R-S8K1-1
- gap:
- recommendation: Providencie esta evidência quando o projeto estiver operacional.

- notes: ['Requisito aplicável apenas em projetos operacionais.', 'Em fase de desenvolvimento/PDD, a evidência operacional ainda não existe — não penaliza o score.']

## Item 45

- requirement\_id: R-CXEP-0
- module: appendix
- title: Biochar characterization standards listed
- status: not\_applicable
- risk: none
- score: nan
- confidence: 1.0
- project\_evidence:
- methodology\_basis: Module: appendix | Requirement ID: R-CXEP-0
- gap:
- recommendation: Providencie esta evidência quando o projeto estiver operacional.
- notes: ['Requisito aplicável apenas em projetos operacionais.', 'Em fase de desenvolvimento/PDD, a evidência operacional ainda não existe — não penaliza o score.']

## Item 46

- requirement\_id: R-7W1N-0
- module: appendix
- title: Biochar physical properties measured
- status: not\_applicable
- risk: none
- score: nan
- confidence: 1.0
- project\_evidence:
- methodology\_basis: Module: appendix | Requirement ID: R-7W1N-0
- gap:
- recommendation: Providencie esta evidência quando o projeto estiver operacional.
- notes: ['Requisito aplicável apenas em projetos operacionais.', 'Em fase de desenvolvimento/PDD, a evidência operacional ainda não existe — não penaliza o score.']

## Item 47

- requirement\_id: R-VGXA-0
- module: appendix
- title: Biochar chemical properties measured
- status: not\_applicable
- risk: none
- score: nan
- confidence: 1.0
- project\_evidence:
- methodology\_basis: Module: appendix | Requirement ID: R-VGXA-0
- gap:
- recommendation: Providencie esta evidência quando o projeto estiver operacional.
- notes: ['Requisito aplicável apenas em projetos operacionais.', 'Em fase de desenvolvimento/PDD, a evidência operacional ainda não existe — não penaliza o score.']

## Item 48

- requirement\_id: R-2TMM-0
- module: appendix

- title: Analytical laboratory identified and qualified
- status: not\_applicable
- risk: none
- score: nan
- confidence: 1.0
- project\_evidence:
- methodology\_basis: Module: appendix | Requirement ID: R-2TMM-0
- gap:
- recommendation: Providencie esta evidência quando o projeto estiver operacional.
- notes: ['Requisito aplicável apenas em projetos operacionais.', 'Em fase de desenvolvimento/PDD, a evidência operacional ainda não existe — não penaliza o score.']